audio.dev
CONFERENCE
BRISTOL & ONLINE 2026

Don't Break the Session

Modernizing a Plugin Back-Catalogue Without Wrecking Saved Projects

Bence Kovács

After an update, a saved session must load and sound the same

After an update, a saved session must load and sound the same

I just upgraded [a plugin] 3. to 3.3.1 and lost ALL my settings/presets I programmed on projects the past year.

After an update, a saved session must load and sound the same

I just upgraded [a plugin] 3. to 3.3.1 and lost ALL my settings/presets I programmed on projects the past year.

The [product] has been [...] overhauled, replacing the names, and more importantly, the SOUNDS of the presets.

After an update, a saved session must load and sound the same

I just upgraded [a plugin] 3. to 3.3.1 and lost ALL my settings/presets I programmed on projects the past year.

The [product] has been [...] overhauled, replacing the names, and more importantly, the SOUNDS of the presets.

that's why you do not update your music DAWs and VST, never.

Ampknobs: one platform, many released products























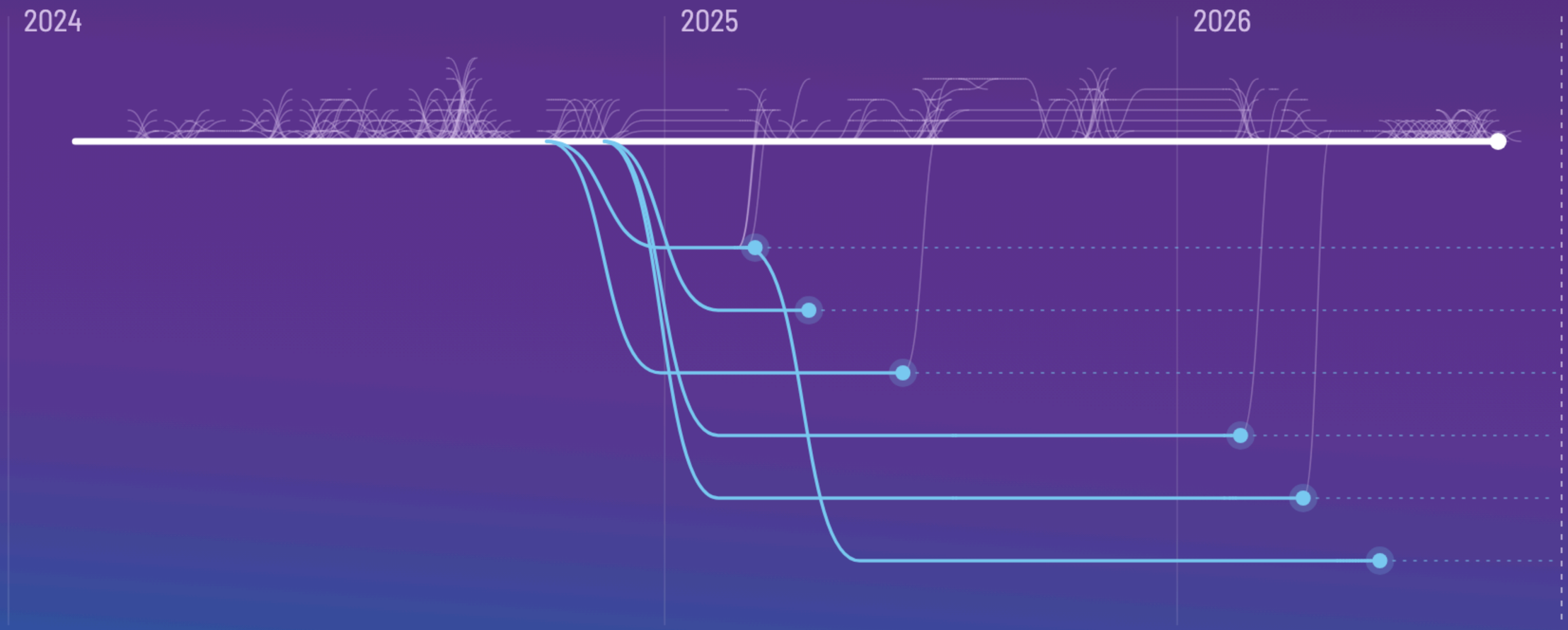




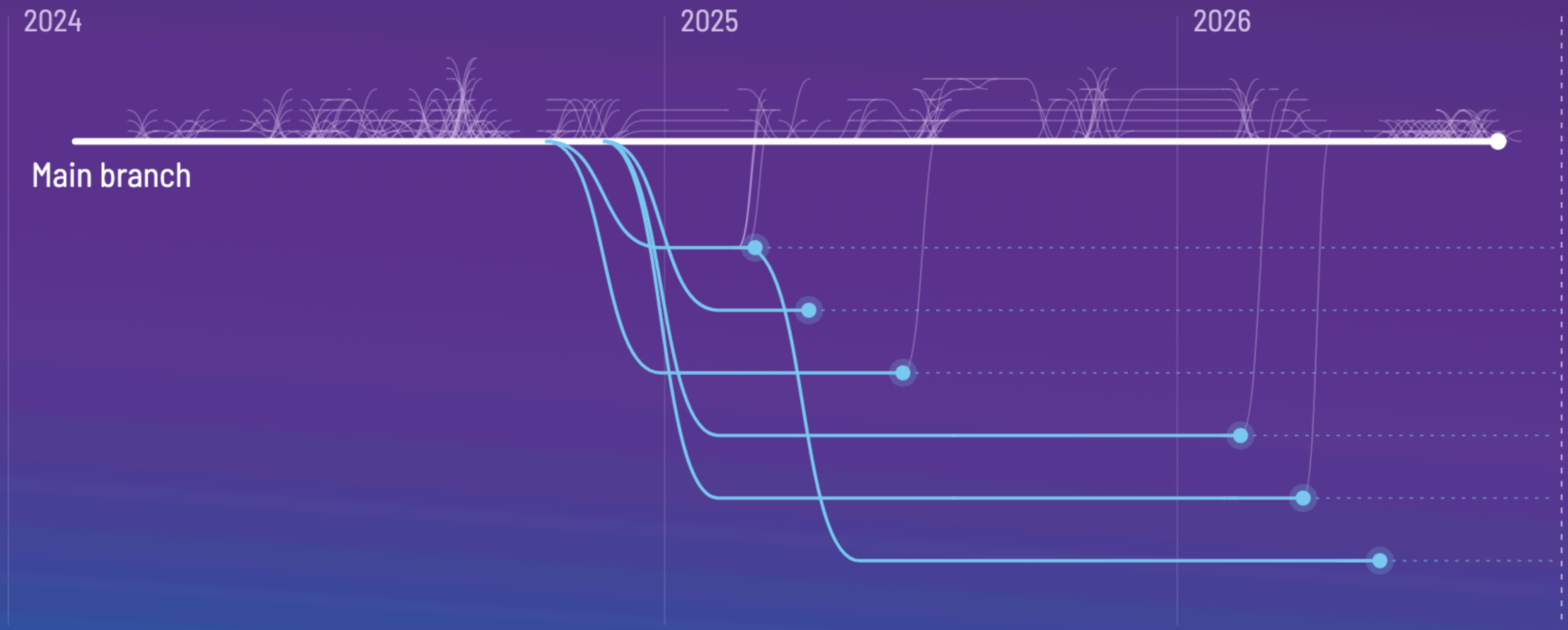




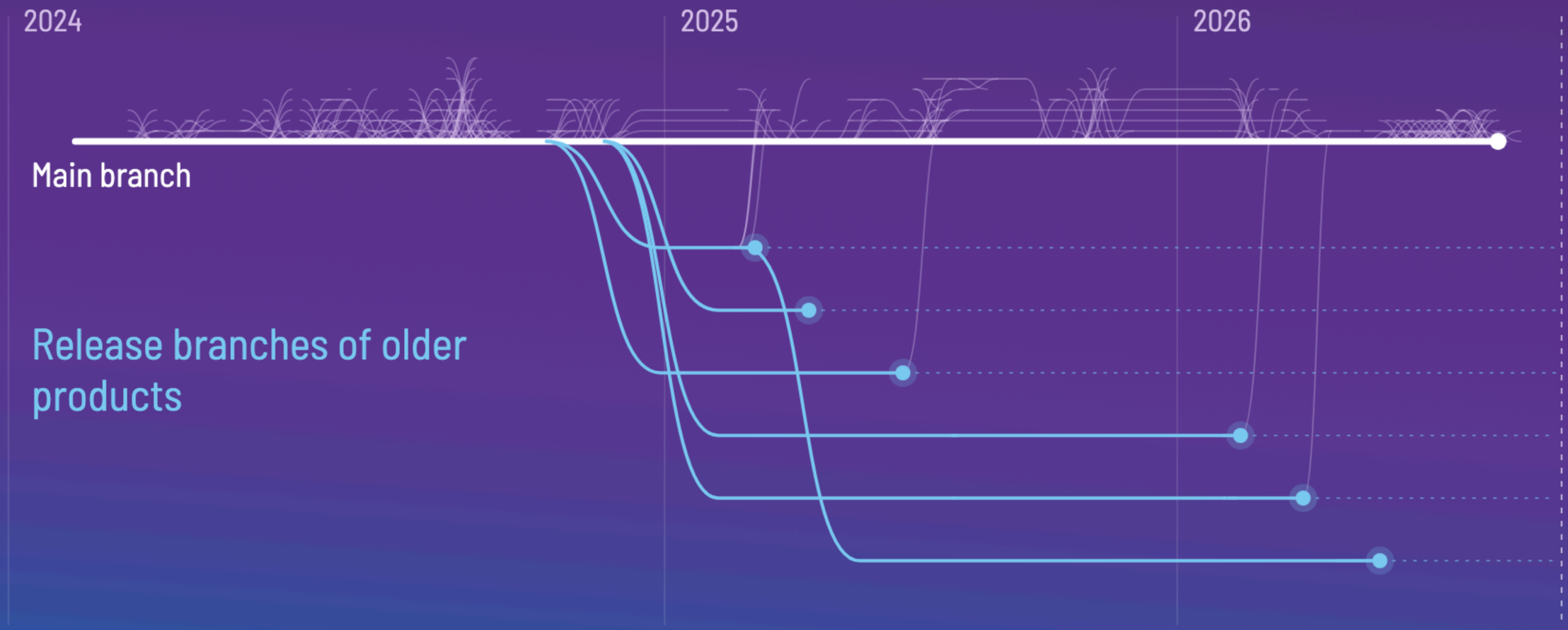
A shared core under every product



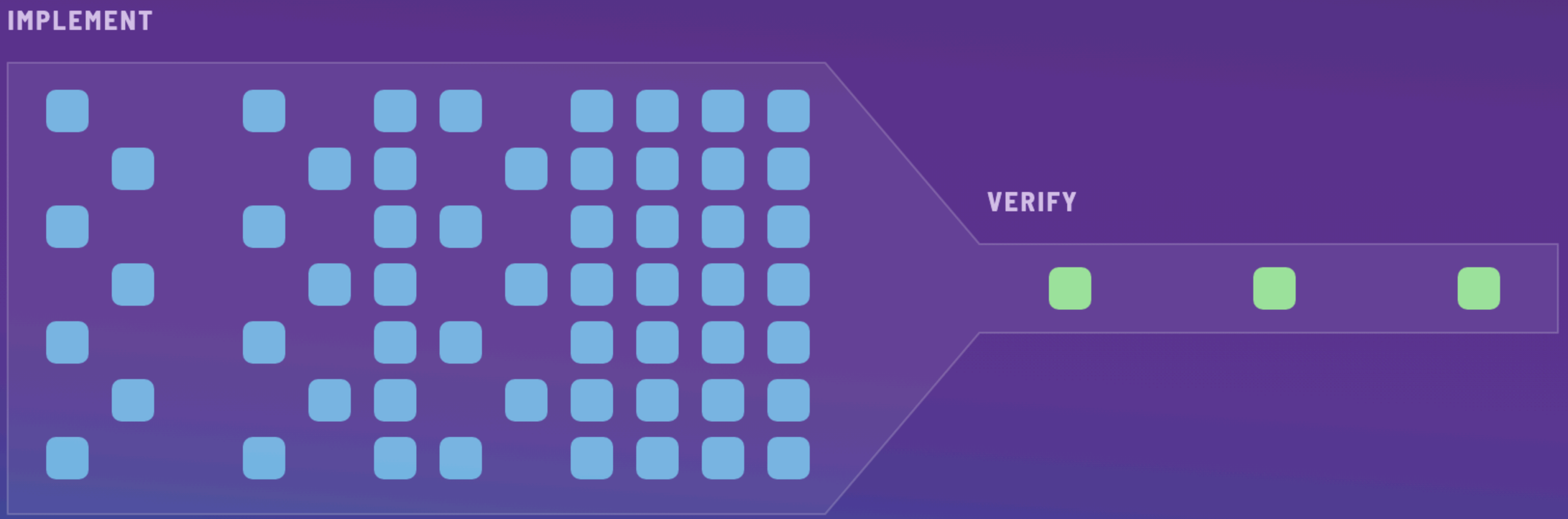
A shared core under every product



A shared core under every product



Generating code became cheap. How about QA?



Give the agent your senses

Agent

E2E HARNESS

See

screenshots · UI state

Act

set any control by name · send MIDI

Hear

capture the output

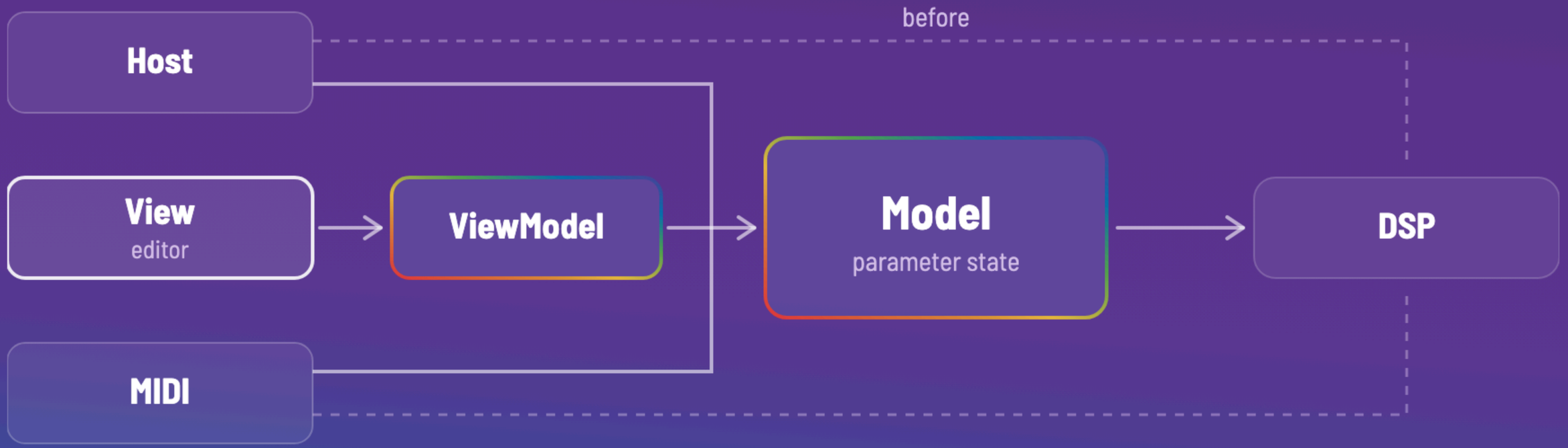


JUCE end to end

https://github.com/FocusriteGroup/juce-end-to-end



Refactoring a legacy plugin



Comparisons with the old build started failing

Agent

WHEN THE EDITOR OPENS

```
setParameter(parVelToVol,
  esnVelToVol / 100.0f);
```

Parameter value gets converted to an int %

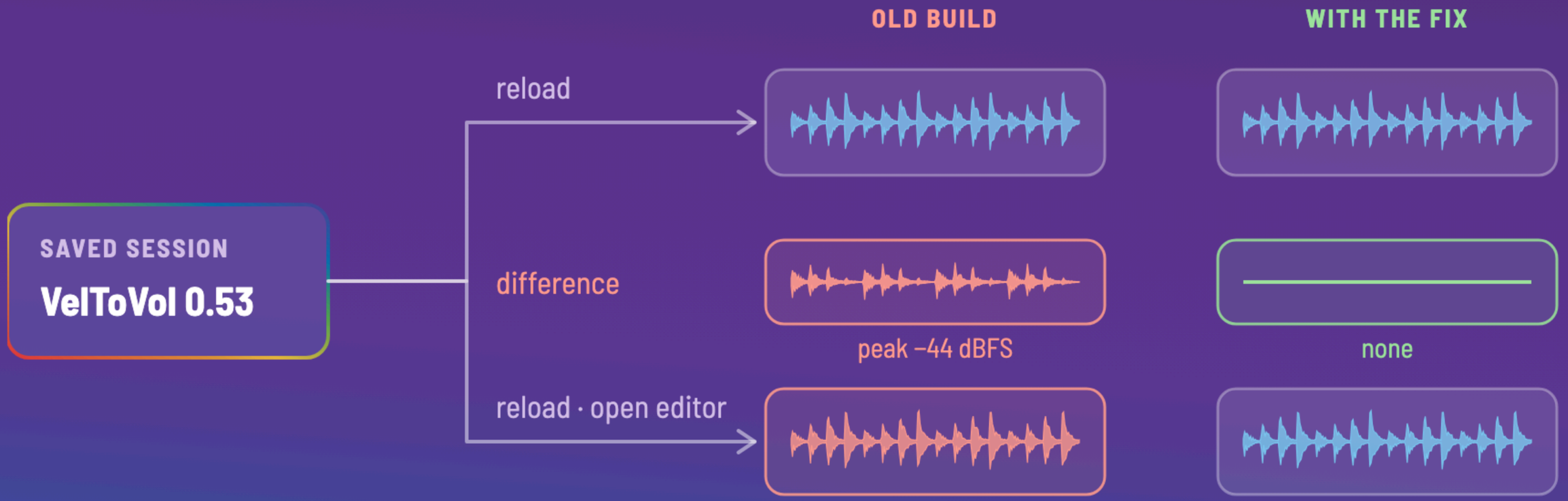
PROBE: EVERY VALUE 0–200 %



Rounding errors detected

```
int(100 * 0.53f) = int(52.999996) = 52
```

The bug and the fix, confirmed in the real plugin



Check that old sessions still work with these three steps

Check that old sessions still work with these three steps

Preserve

renders of the
released build

Check that old sessions still work with these three steps

Preserve

renders of the
released build

Verify

every route
that can break

Check that old sessions still work with these three steps

Preserve

renders of the released build

Verify

every route that can break

Break it

on purpose and confirm the test catches it

Preserve

renders of the released build

Verify

every route that can break

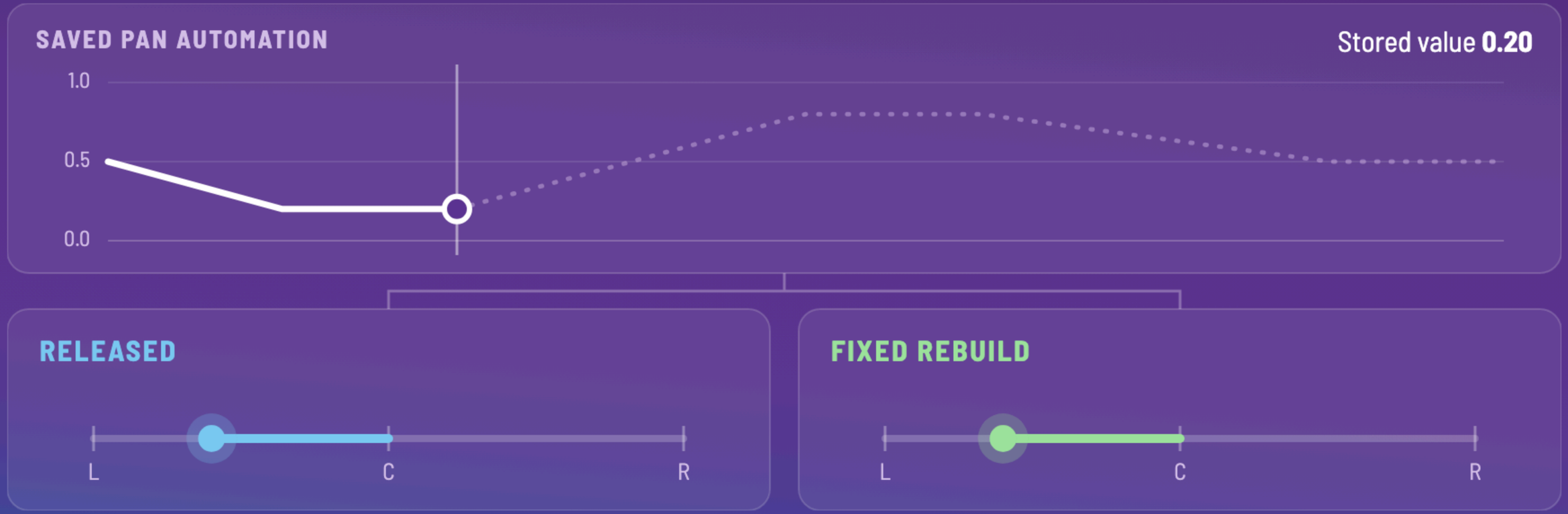
Break it

on purpose and confirm the test catches it

Skeletons in the shared code



The fixed rebuild pans saved automation the way the release did



Which contracts to keep after an update?

WHAT MUST STAY THE SAME	HOW CLOSELY IT MUST MATCH
Plugin identity & parameter meaning	Exactly the same
Session state, presets & settings	Reloads to the same settings
Content & file references	The same files load and play
Host integration, routing & timing	The host sees the same behavior
UI behavior & accessibility	Same actions, same results
Lifecycle & error handling	Starts, stops and fails the same way
Installation, upgrades & licensing	Existing installs keep working
Audio output	Bit-identical, or within a set tolerance
UI appearance	Looks the same at every supported UI scale
Performance & resources	Within an agreed CPU and memory budget

Record how the old build responds to known inputs

INPUT		OLD BUILD RESPONDED WITH
A saved session plays a MIDI groove	→	The audio it rendered
A preset is loaded and saved again	→	The state it saved
Mute and Solo are clicked in the mixer	→	The screenshot and control values
A QA test scenario runs	→	The test result

Record how the old build responds to known inputs

INPUT		OLD BUILD RESPONDED WITH
A saved session plays a MIDI groove	→	The audio it rendered
A preset is loaded and saved again	→	The state it saved
Mute and Solo are clicked in the mixer	→	The screenshot and control values
A QA test scenario runs	→	The test result

The agent plays the same inputs into the new build and compares its responses with these.

Record how the old build responds to known inputs

INPUT		OLD BUILD RESPONDED WITH
A saved session plays a MIDI groove	→	The audio it rendered
A preset is loaded and saved again	→	The state it saved
Mute and Solo are clicked in the mixer	→	The screenshot and control values
A QA test scenario runs	→	The test result

The agent plays the same inputs into the new build and compares its responses with these.

Same content, sample rate, block size and host on both sides.

Preserve

renders of the released build

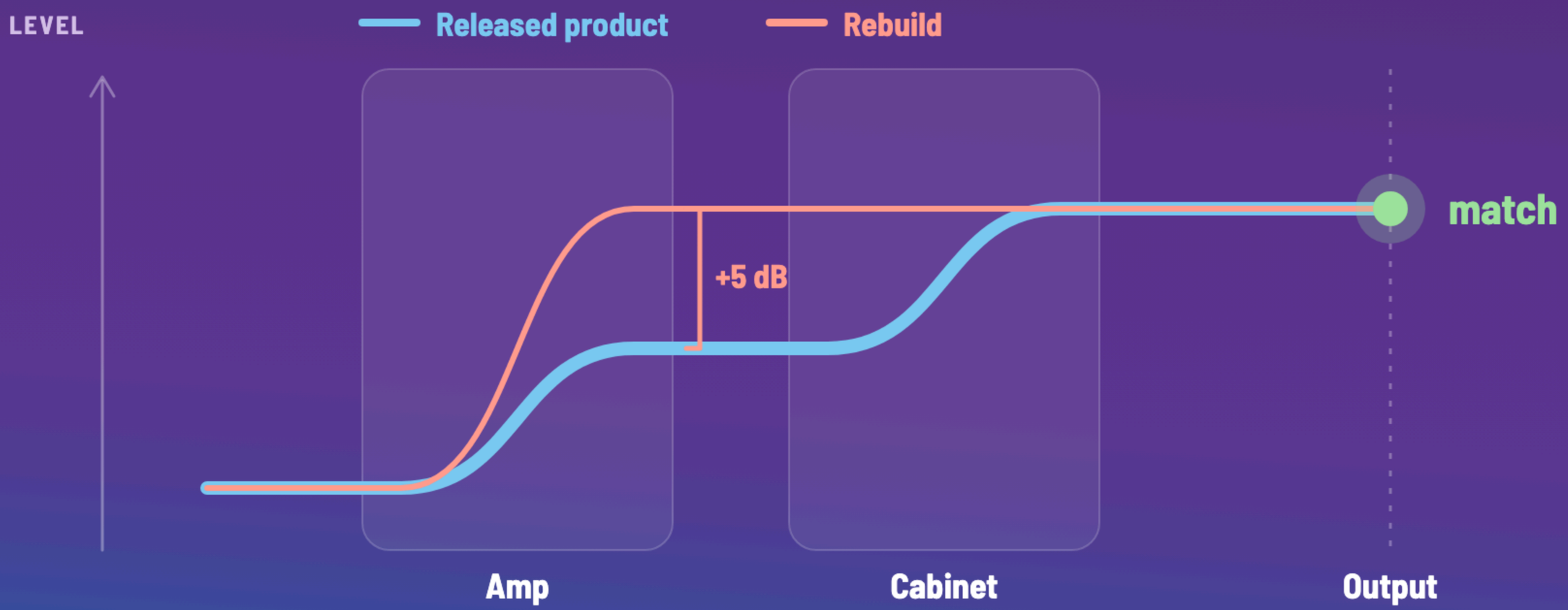
Verify

every route that can break

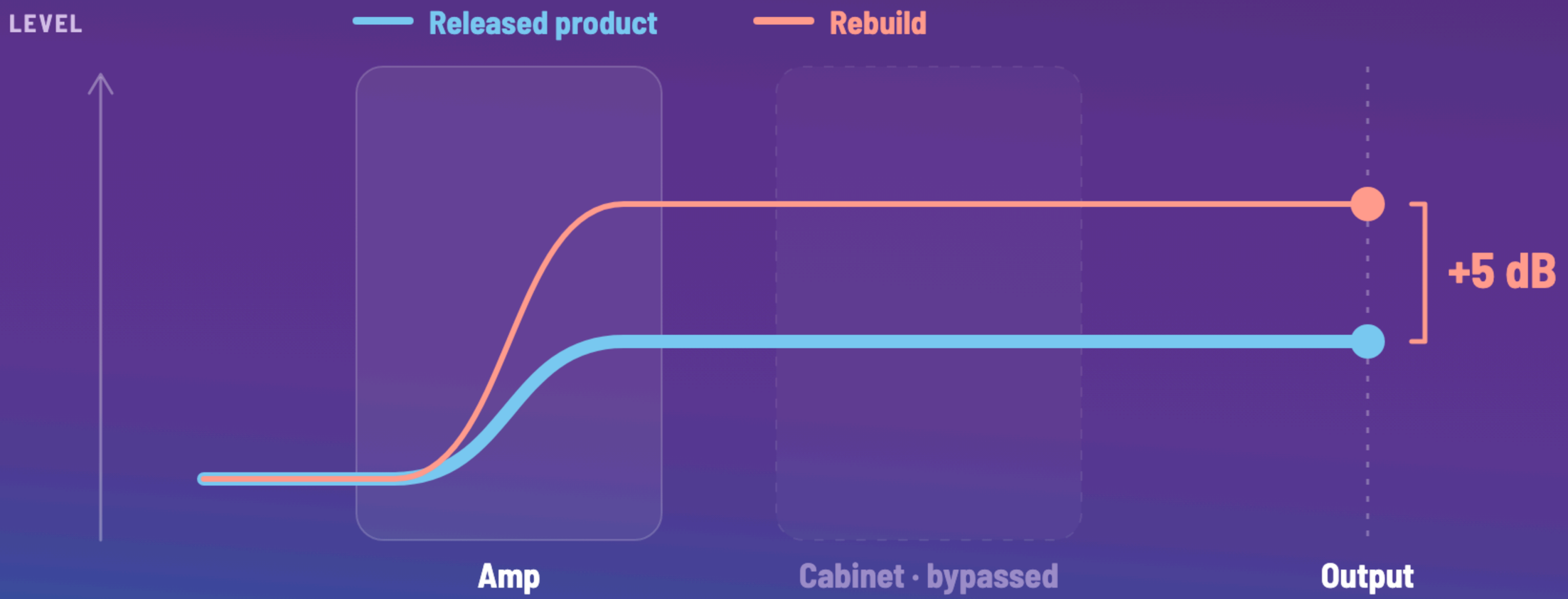
Break it

on purpose and confirm the test catches it

Cab on: the levels match



Cab bypassed: the rebuild plays louder



Preserve

renders of the released build

Verify

every route that can break

Break it

on purpose and confirm the test catches it

Mutation test: break the product on purpose and see which tests fail

BROKEN ON PURPOSE

Outputs 3 and 4 swapped

Mutation test: break the product on purpose and see which tests fail

BROKEN ON PURPOSE

Outputs 3 and 4 swapped

SEVEN RECORDED INPUTS, REPLAYED



6 match the old build
1 differs

Mutation test: break the product on purpose and see which tests fail

BROKEN ON PURPOSE

Outputs 3 and 4 swapped

SEVEN RECORDED INPUTS, REPLAYED



6 match the old build
1 differs

Reloading an edited session

COMPARED WITH THE OLD BUILD

New output 3 = old output 4

New output 4 = old output 3

Mutation test: break the product on purpose and see which tests fail

BROKEN ON PURPOSE

Outputs 3 and 4 swapped

SEVEN RECORDED INPUTS, REPLAYED



6 match the old build
1 differs

Reloading an edited session

COMPARED WITH THE OLD BUILD

New output 3 = old output 4

New output 4 = old output 3

WHAT THE AGENT CONCLUDES

The output routing is swapped

so it reads the routing code

Negative control: for every new test, switch off the feature it tests and watch the test fail

Feature under test: the memory limit for loading drum kits

	NORMAL BUILD	MEMORY LIMIT SWITCHED OFF	
A load at the limit is refused	PASS	FAIL	
Many loads at once stay within the limit	PASS	FAIL	
Samples in use count toward the limit	PASS	FAIL	
Memory use levels off at the limit	PASS	FAIL	
Shutdown frees the memory	PASS	PASS	doesn't depend on the limit

Negative control: for every new test, switch off the feature it tests and watch the test fail

Feature under test: the memory limit for loading drum kits

	NORMAL BUILD	MEMORY LIMIT SWITCHED OFF	
A load at the limit is refused	PASS	FAIL	
Many loads at once stay within the limit	PASS	FAIL	
Samples in use count toward the limit	PASS	FAIL	
Memory use levels off at the limit	PASS	FAIL	
Shutdown frees the memory	PASS	PASS	doesn't depend on the limit

The agent writes the broken build together with the test.

Summary

Summary

- Let agents drive your app and compare the behavior of different versions, just like you would.

Summary

- Let agents drive your app and compare the behavior of different versions, just like you would.
- Incrementally build up your test coverage. Bug fixes should be merged along with new regression tests.

Summary

- Let agents drive your app and compare the behavior of different versions, just like you would.
- Incrementally build up your test coverage. Bug fixes should be merged along with new regression tests.
- Validate that your tests check the right thing and actually catch the damage.

Slides

https://github.com/Bogren-Digital/dont-break-the-session

